# AUTOMATED GLACIAL LAKE DETECTION USING DEEPLABV3+


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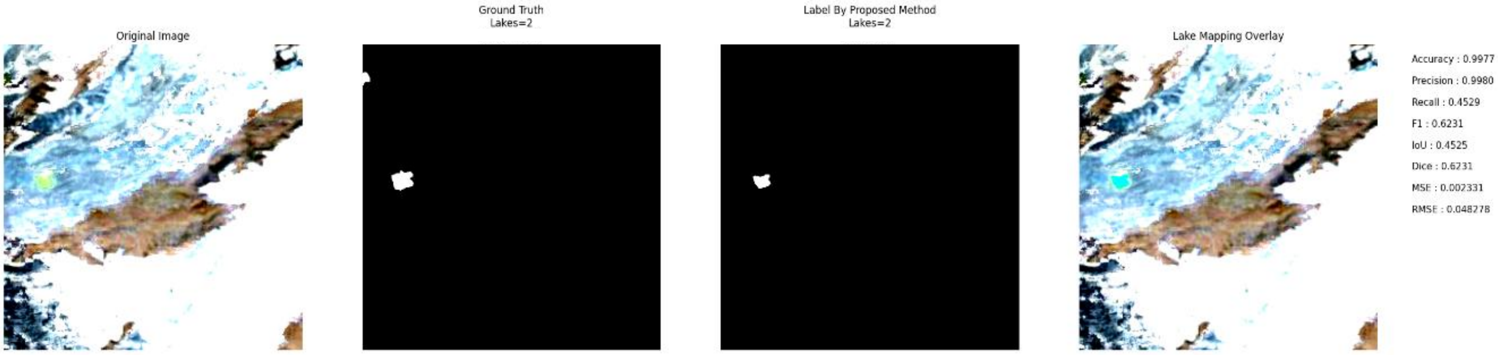
## Abstract

Glacial lakes are rapidly increasing in number and size due to climate change and glacier retreat. These lakes pose a significant risk because they can trigger Glacial Lake Outburst Floods (GLOFs), causing severe damage to downstream communities and infrastructure. Manual mapping of glacial lakes from satellite imagery is labor-intensive and time-consuming. This project presents an automated deep learning framework for glacial lake detection using DeepLabV3+ with an EfficientNet-B4 backbone. The proposed system performs semantic segmentation on satellite imagery to accurately identify lake regions while minimizing confusion with snow, glacier ice, debris-covered ice, and terrain shadows. The model achieved an Accuracy of 99.62%, Precision of 98.09%, Recall of 93.49%, F1 Score of 95.74%, IoU of 91.82%, and Cohen's Kappa of 0.955.

## Class-wise Output

## 1. Cloud Cover

**Description:** Output images obtained for cloud-covered regions

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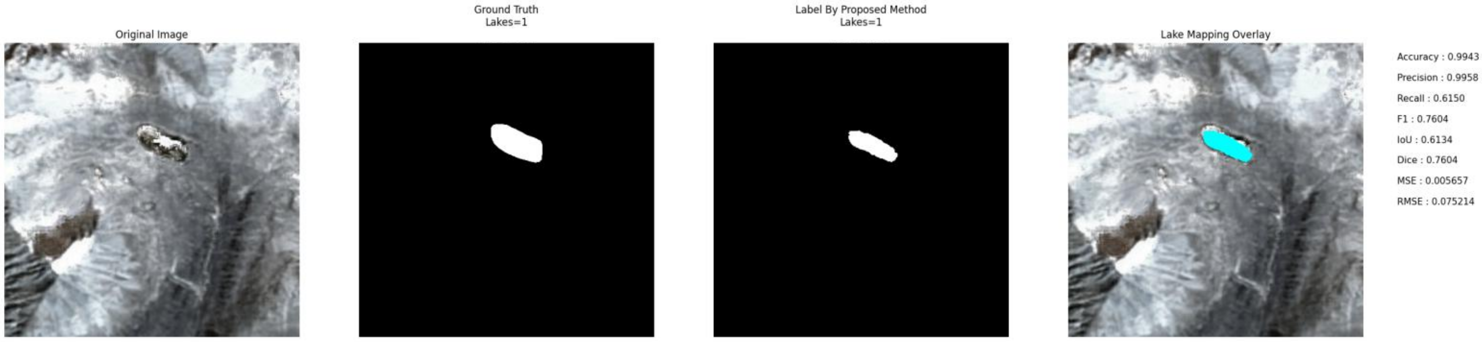


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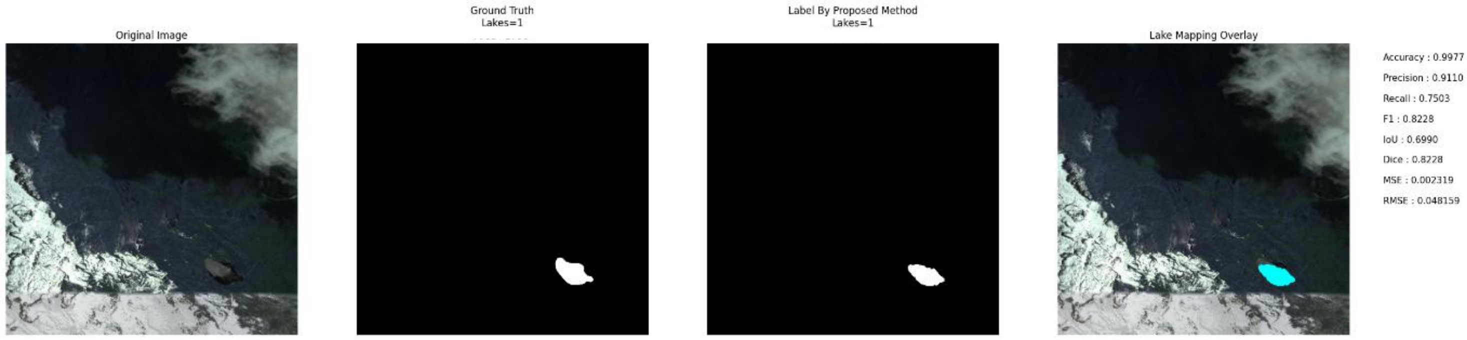


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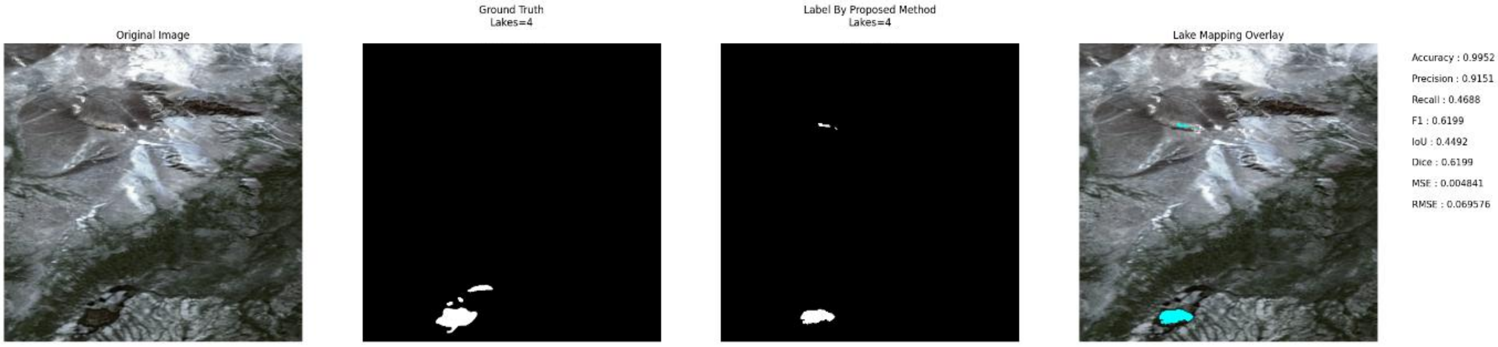


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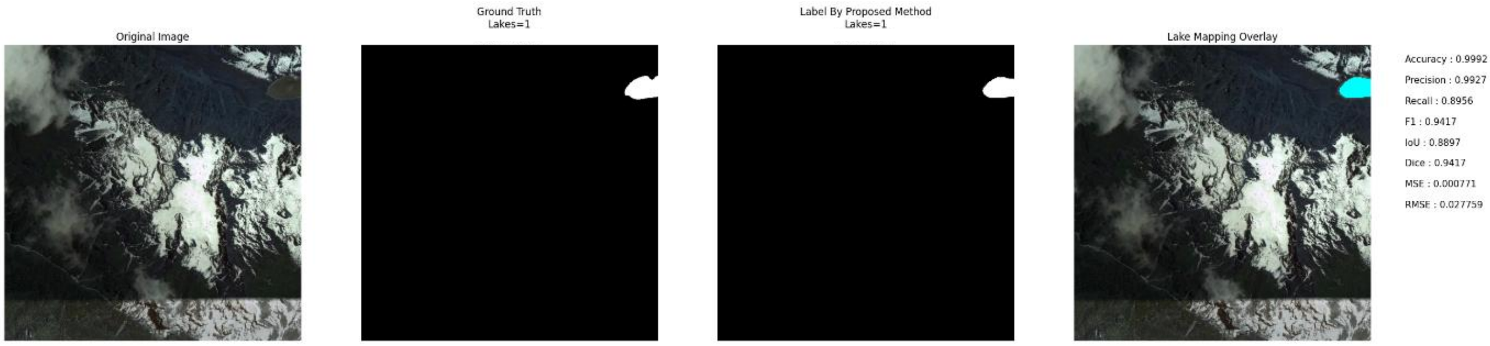


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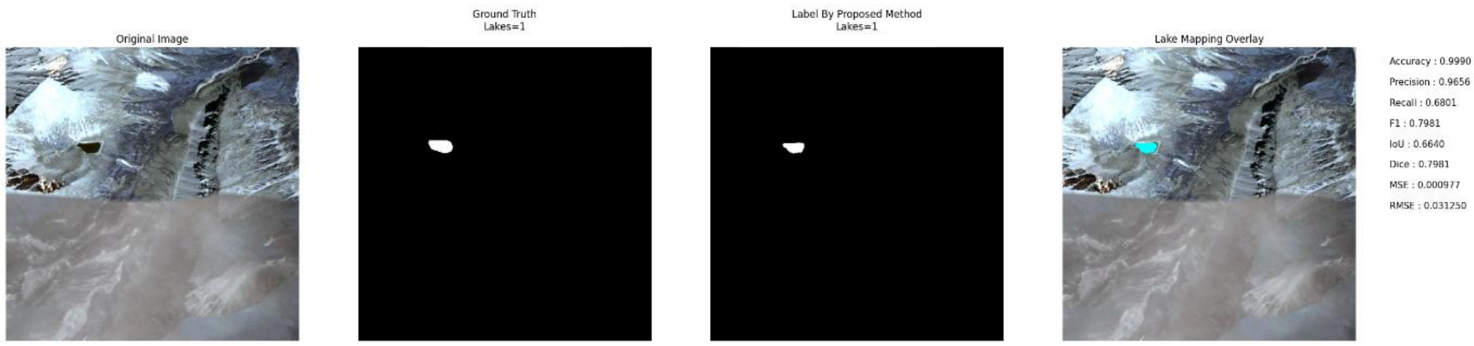


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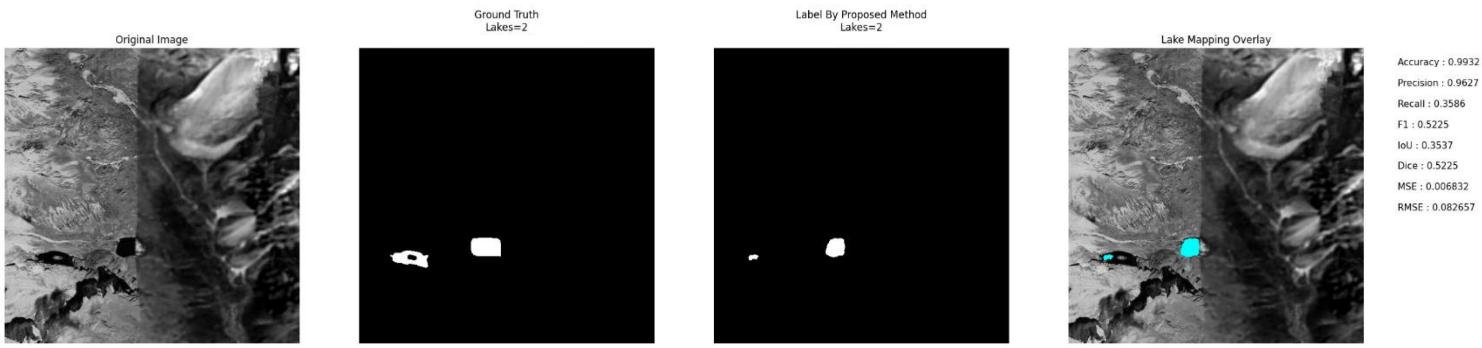


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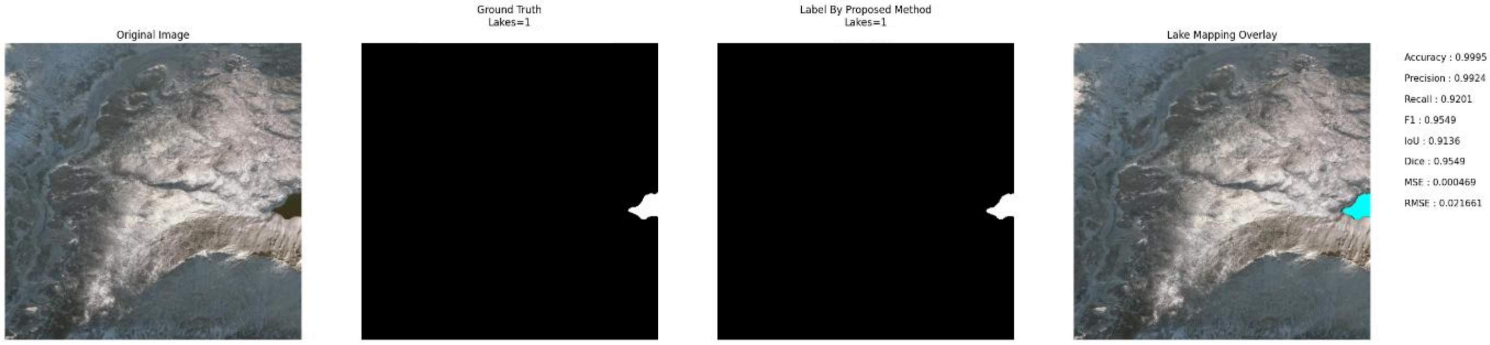


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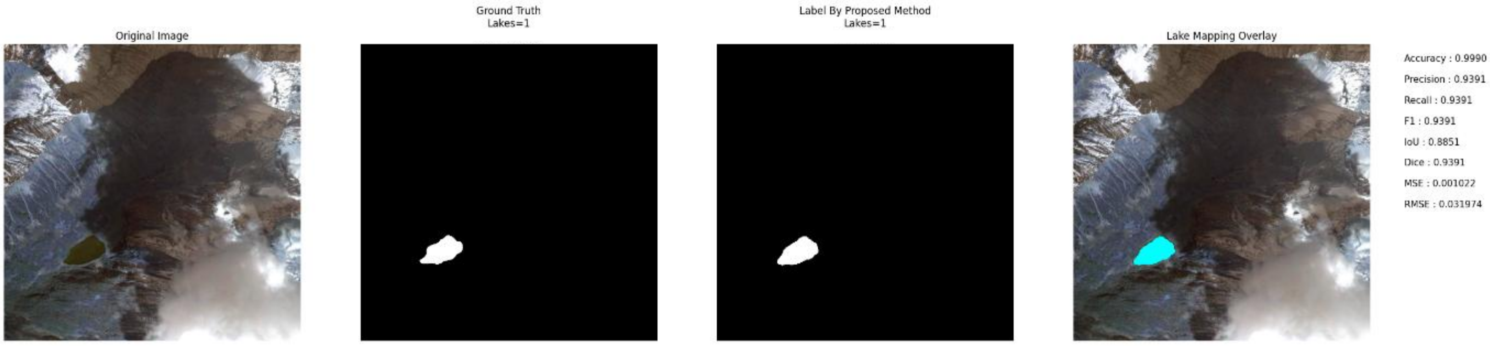


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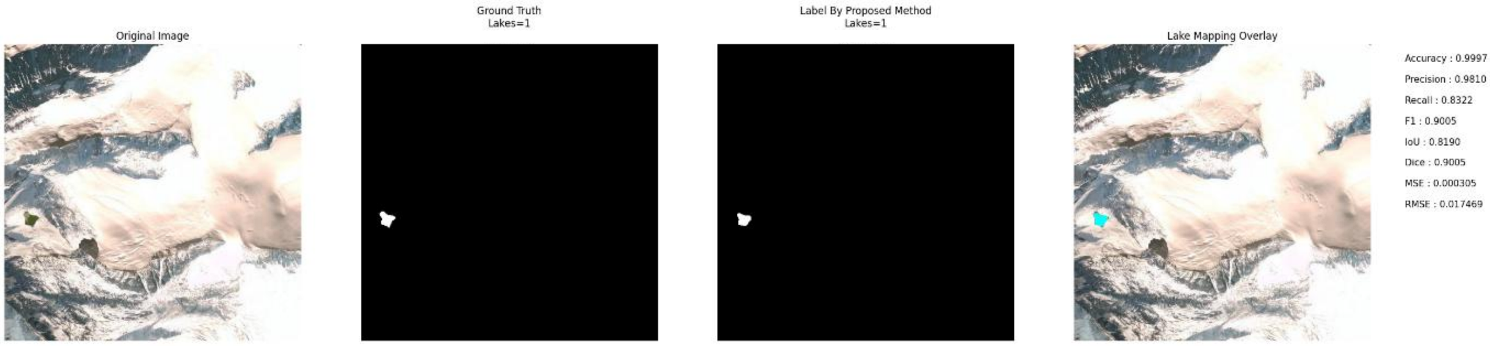


Image No: 74

## 2. Snow Cover

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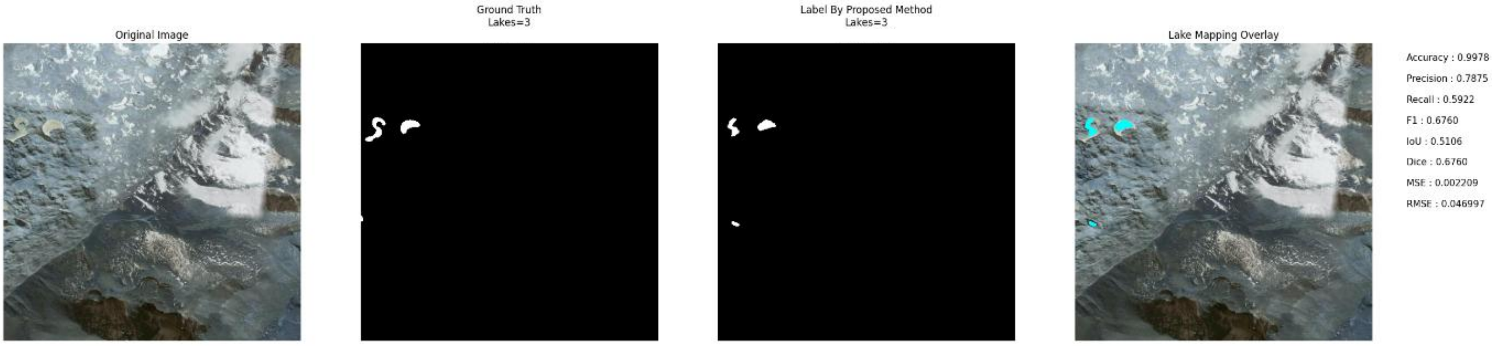


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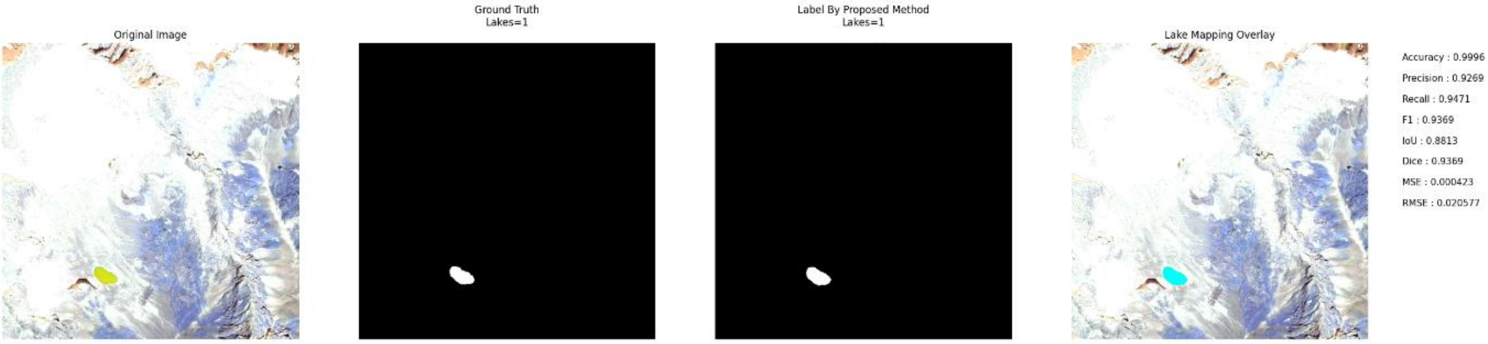


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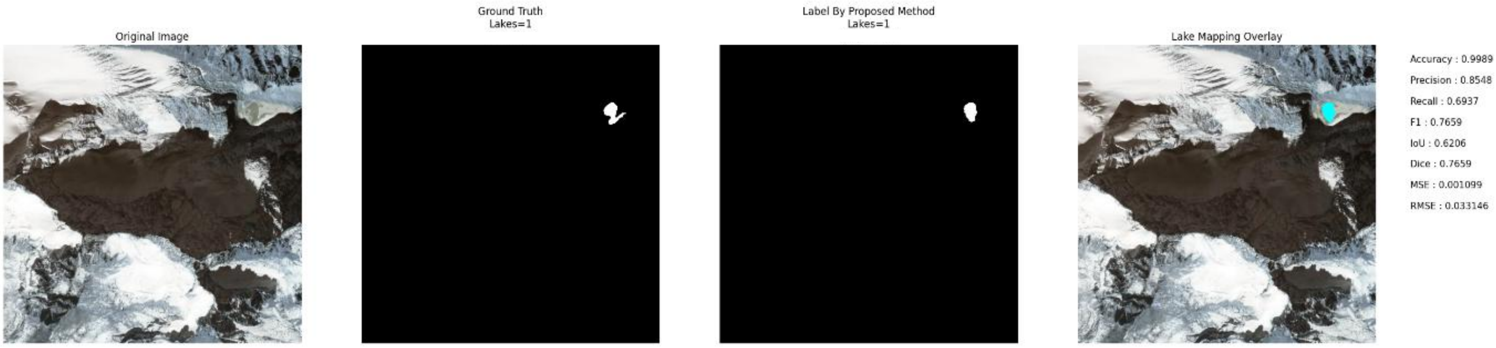


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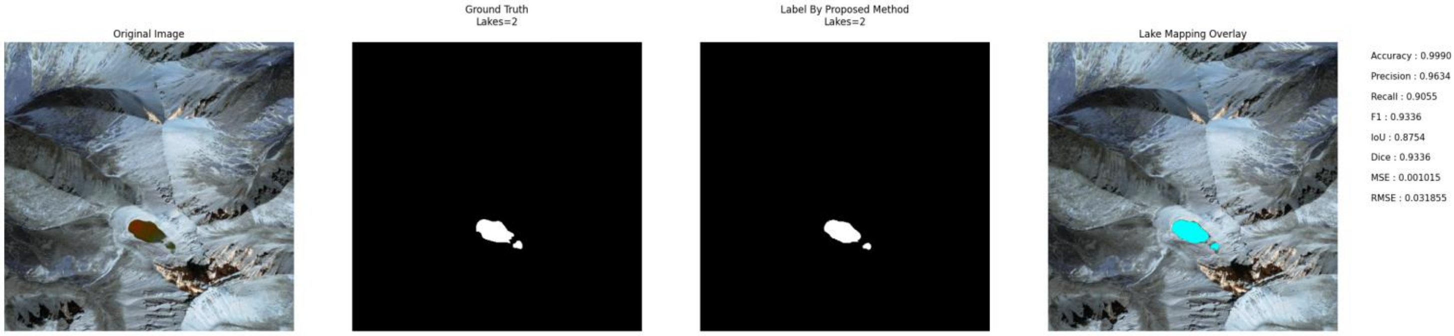


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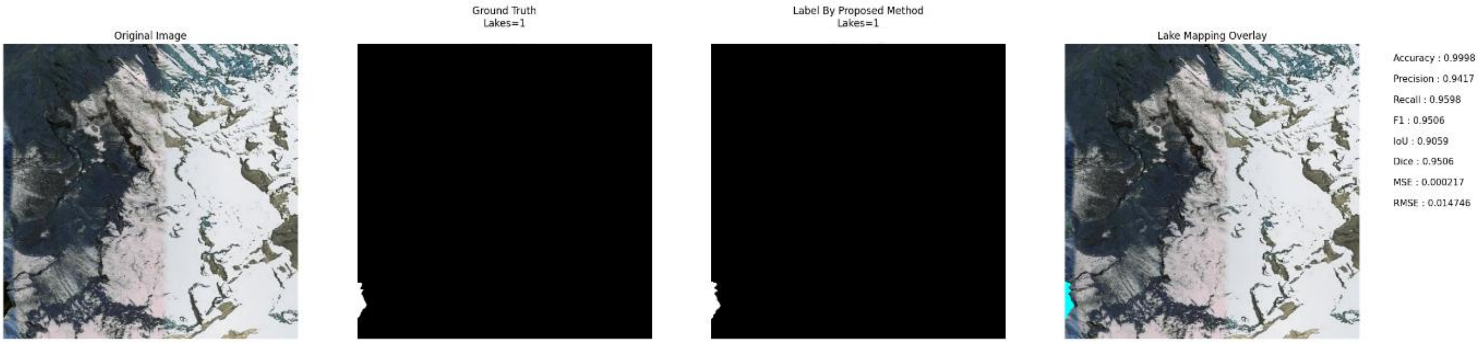


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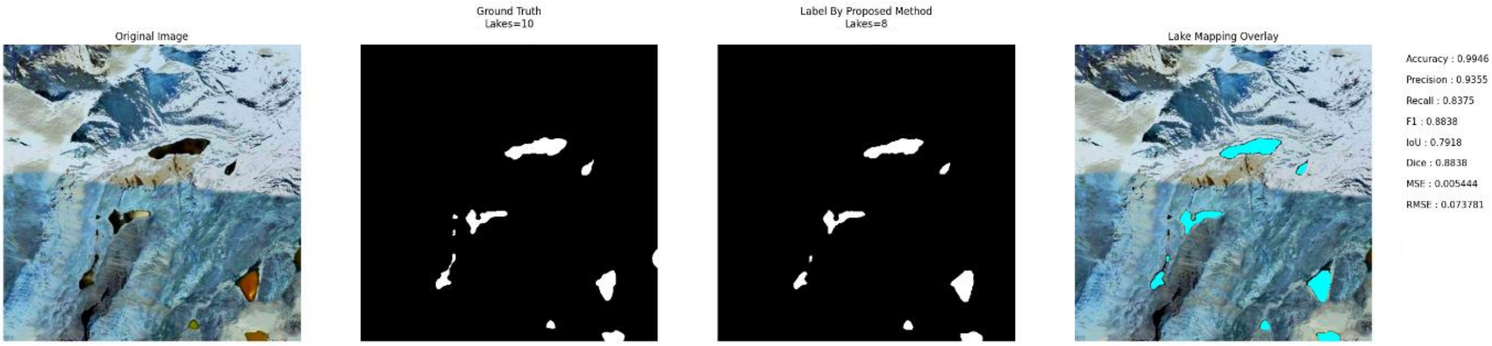


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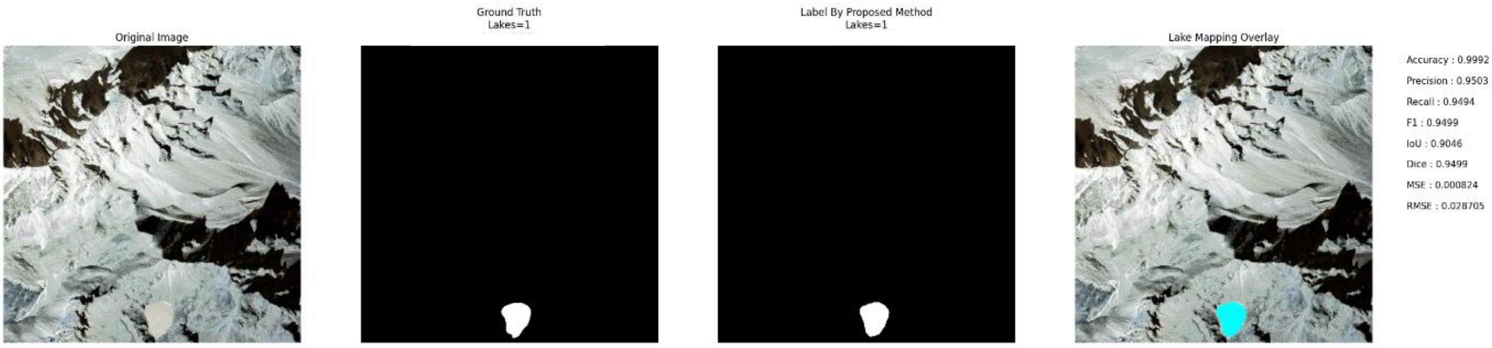


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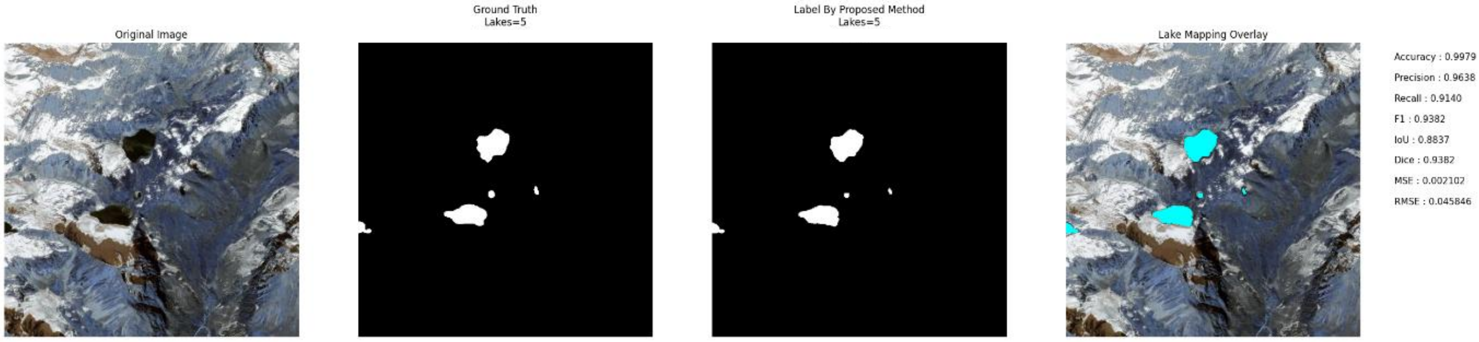


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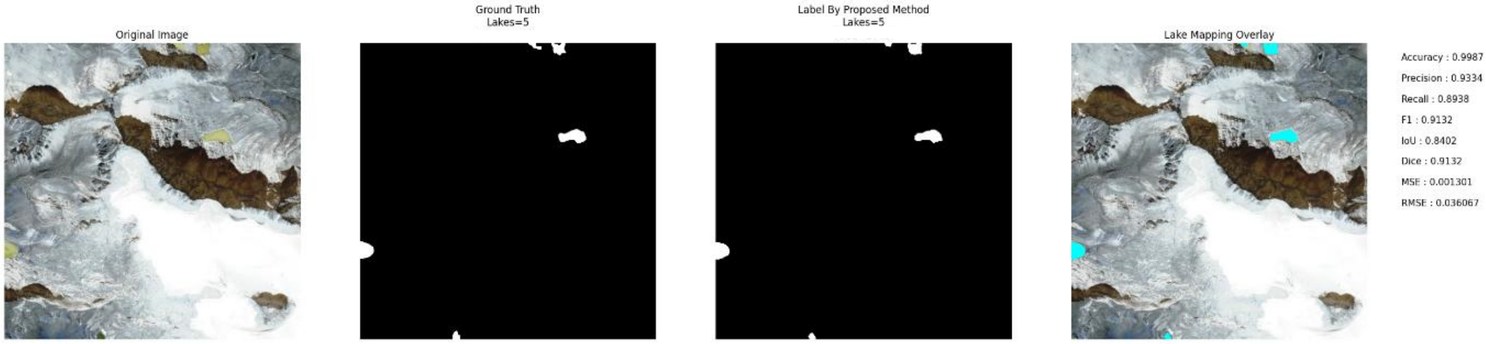


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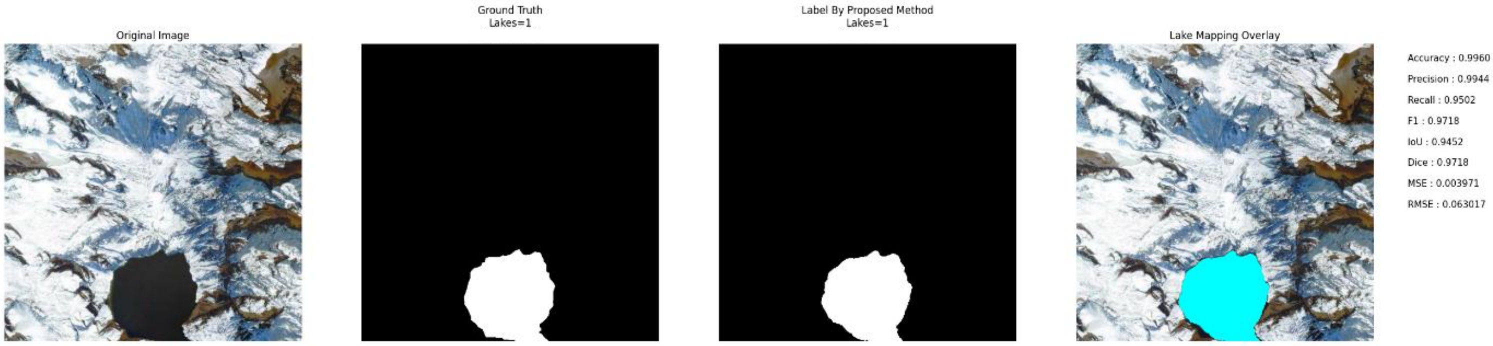


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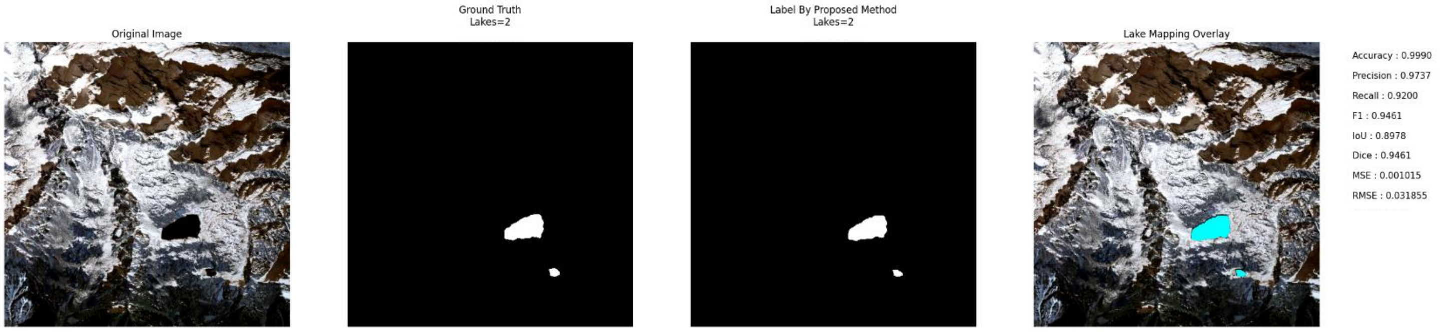
## 3. Terrain Shadow

**Description:** Output images obtained for terrain shadow regions.

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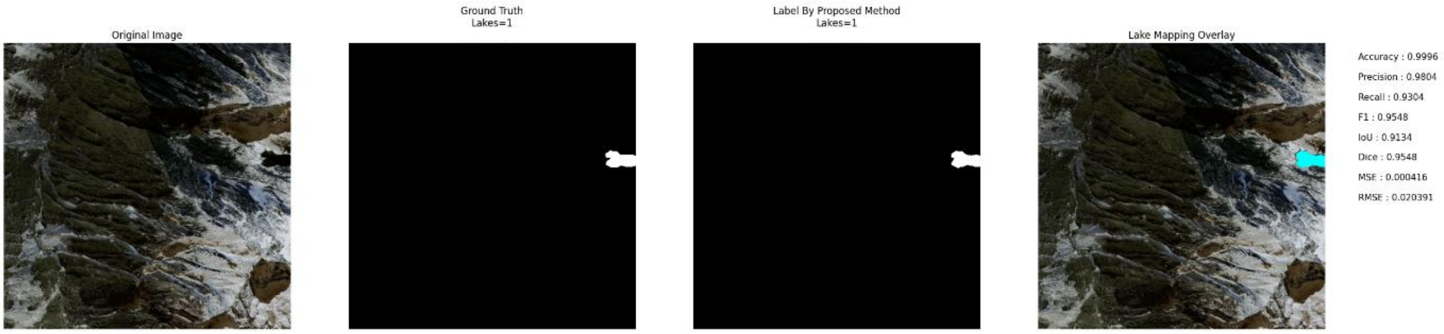


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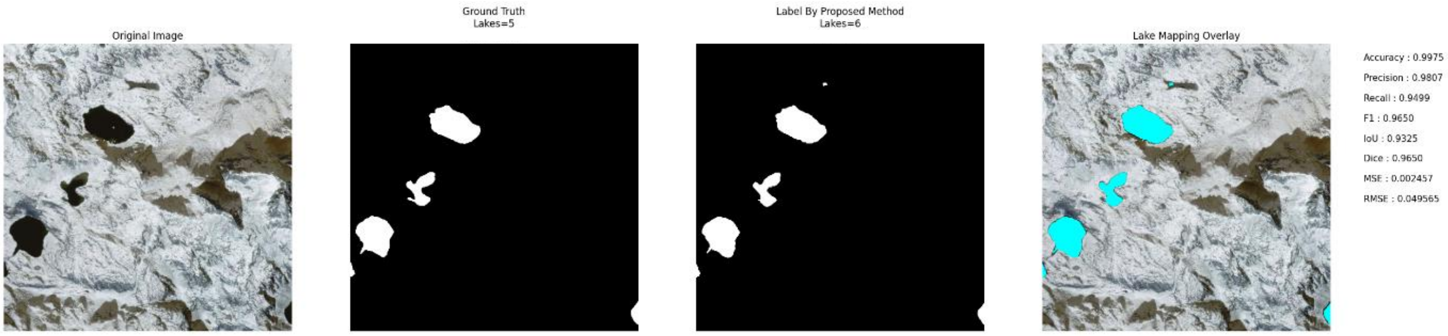


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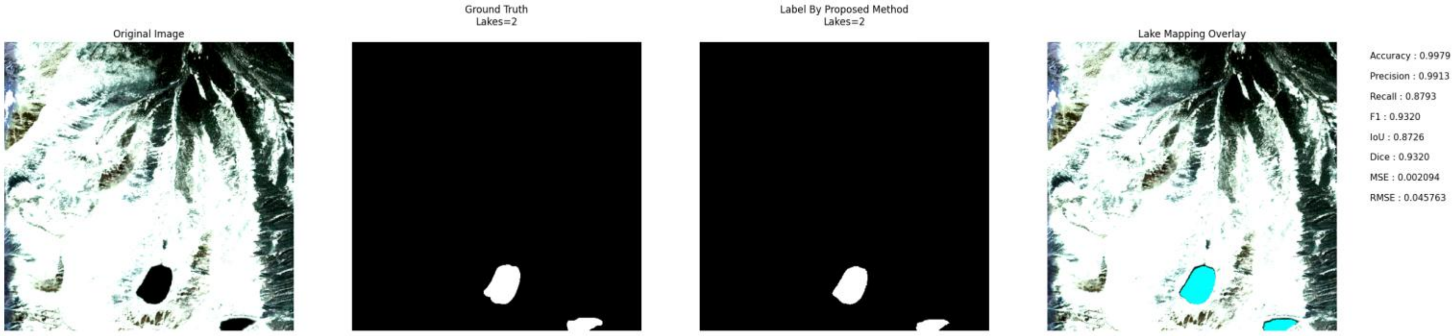


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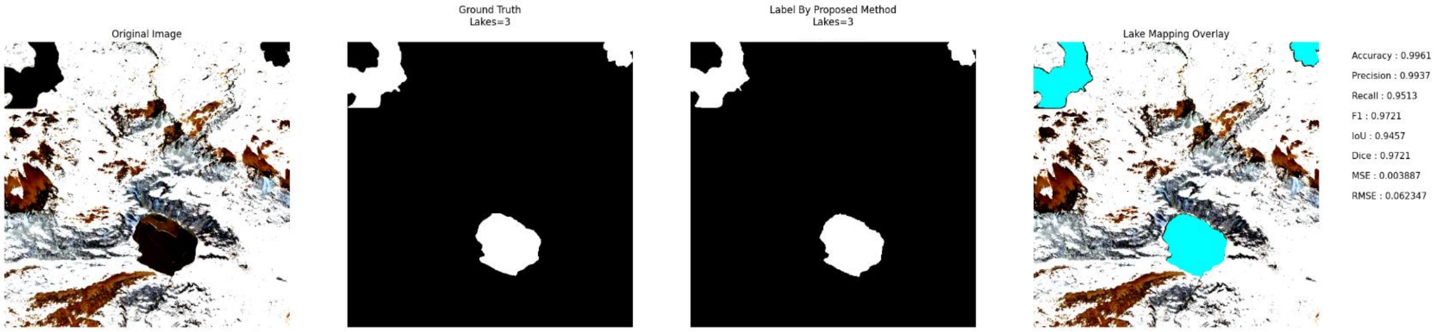


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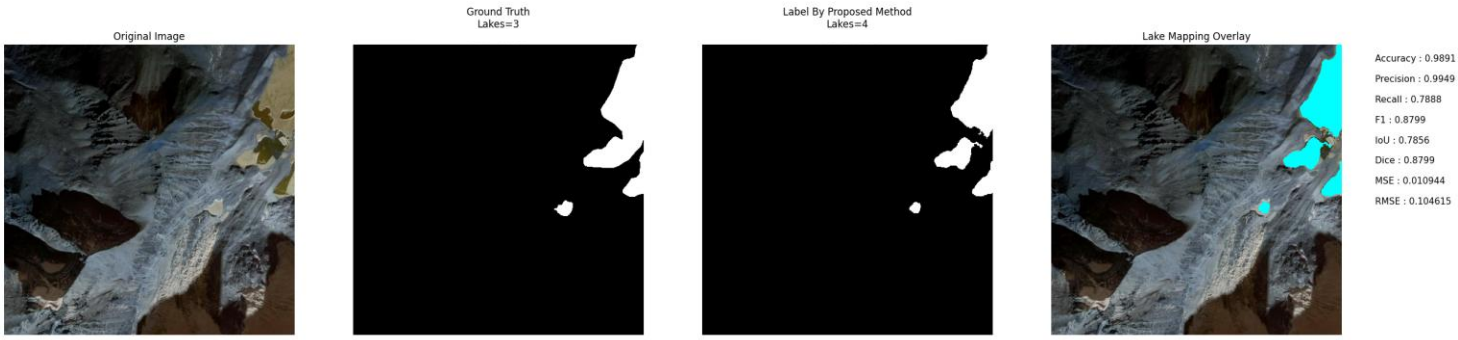


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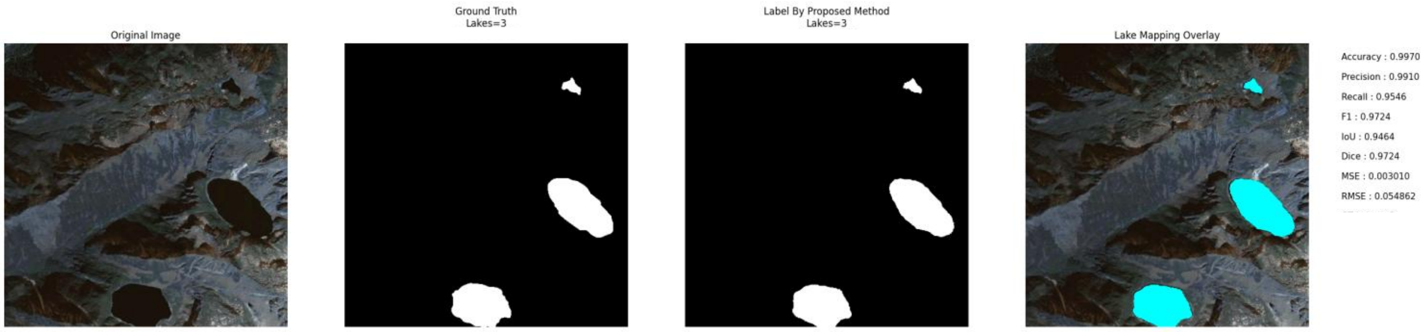


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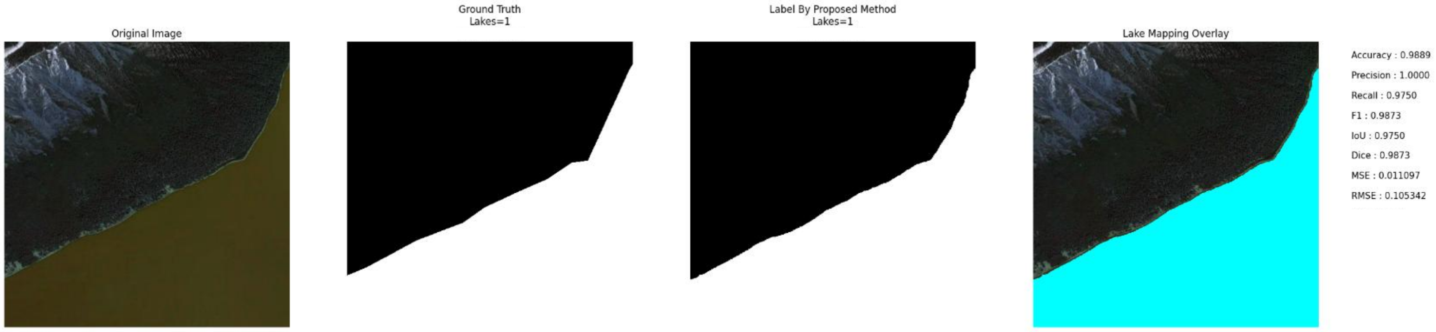


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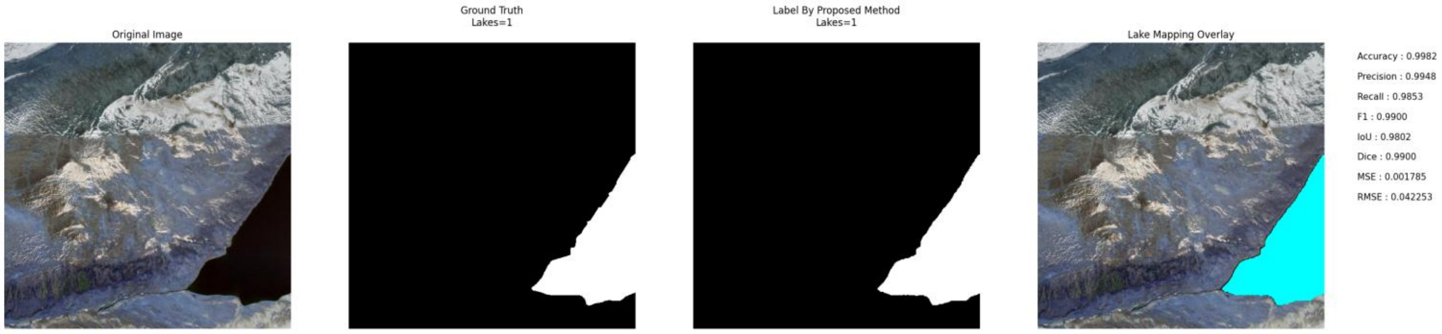


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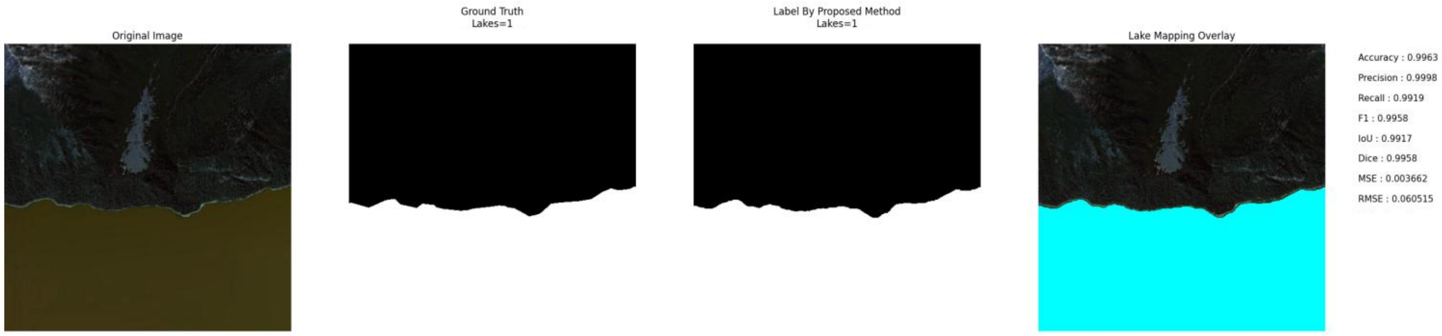


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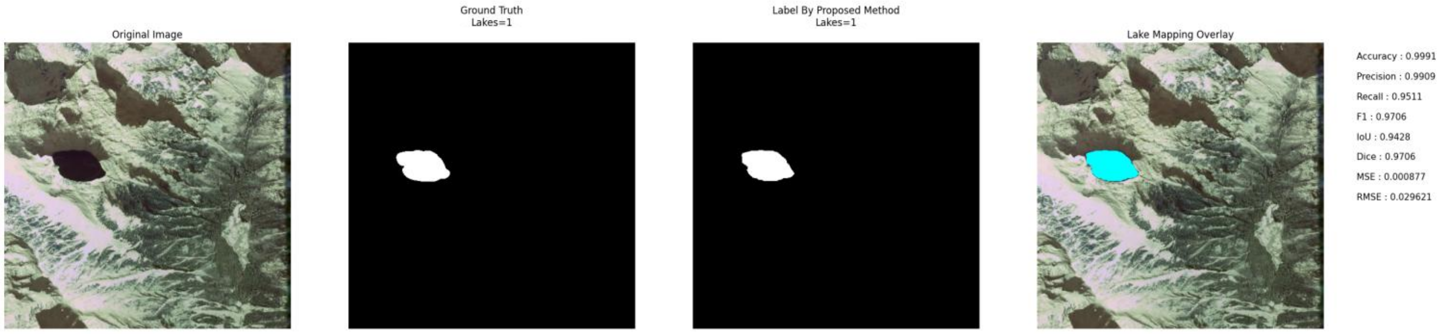
## 4. Debris Cover

**Description:** Output images obtained for debris-covered regions.

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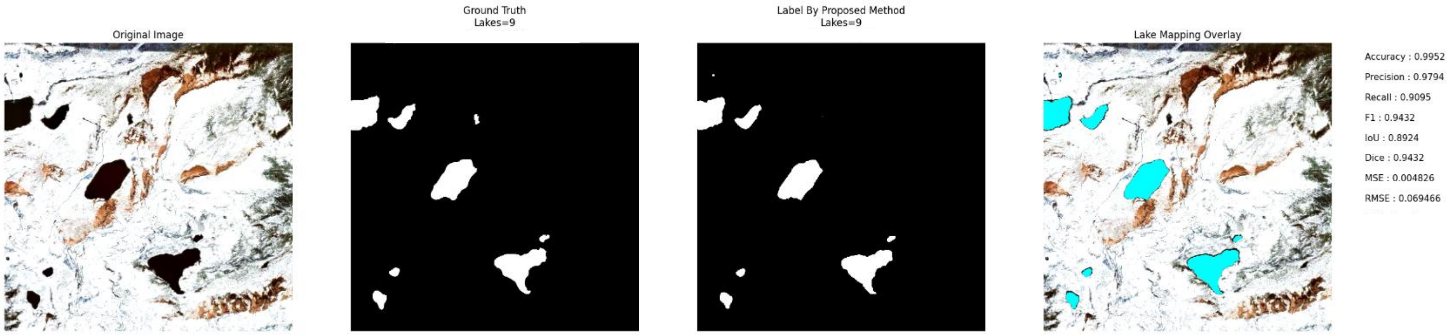


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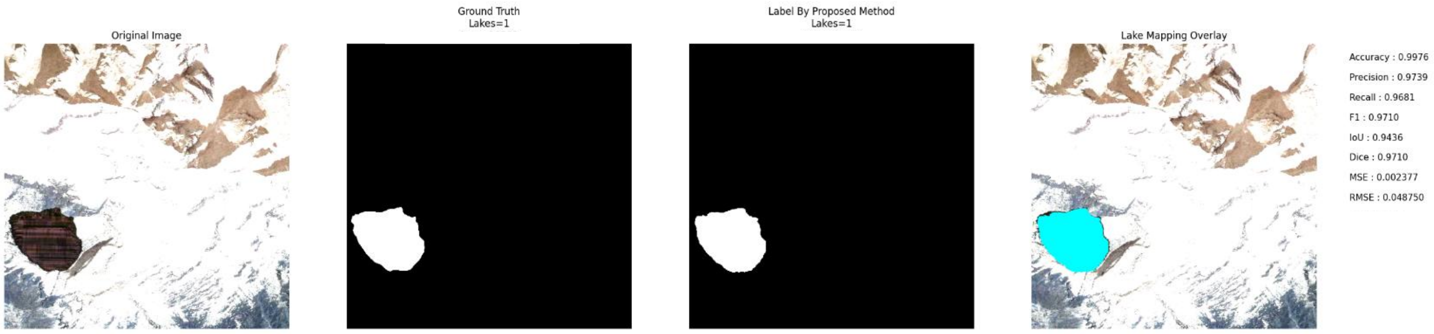


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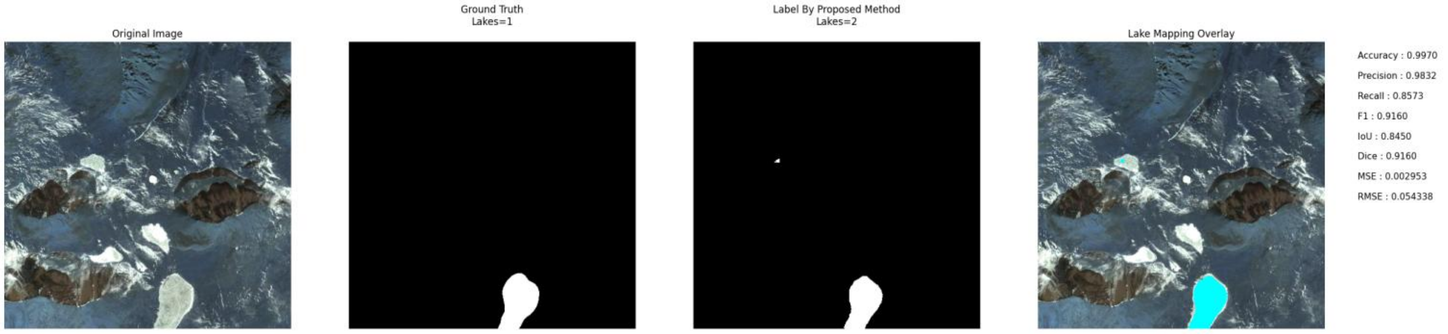


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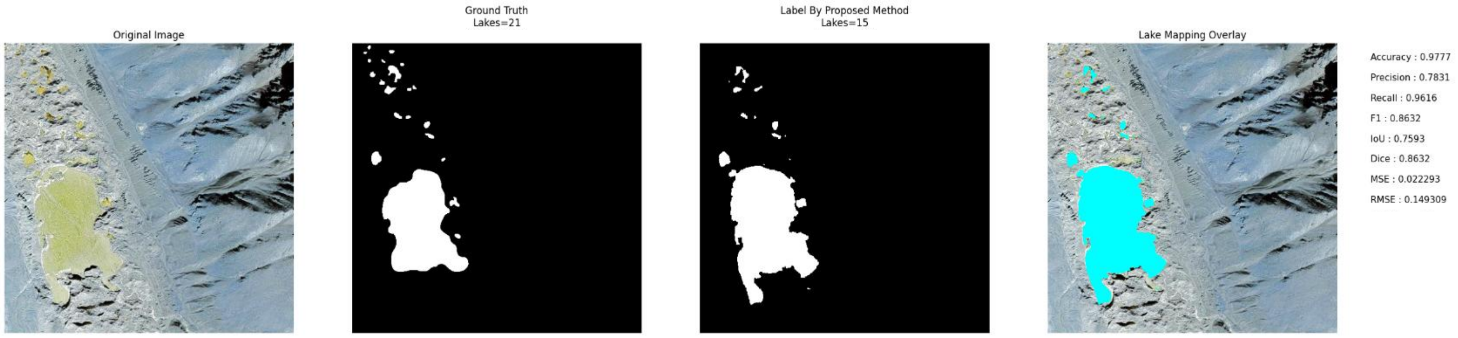


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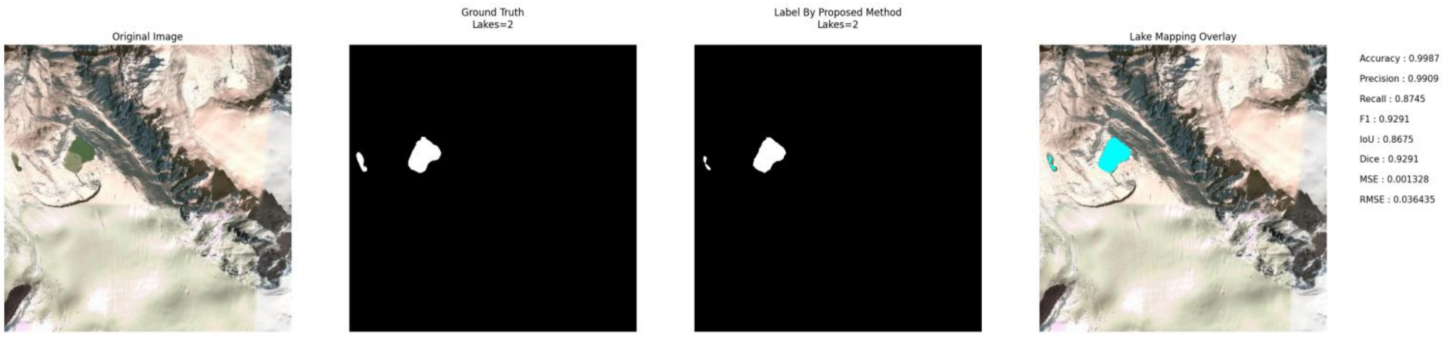


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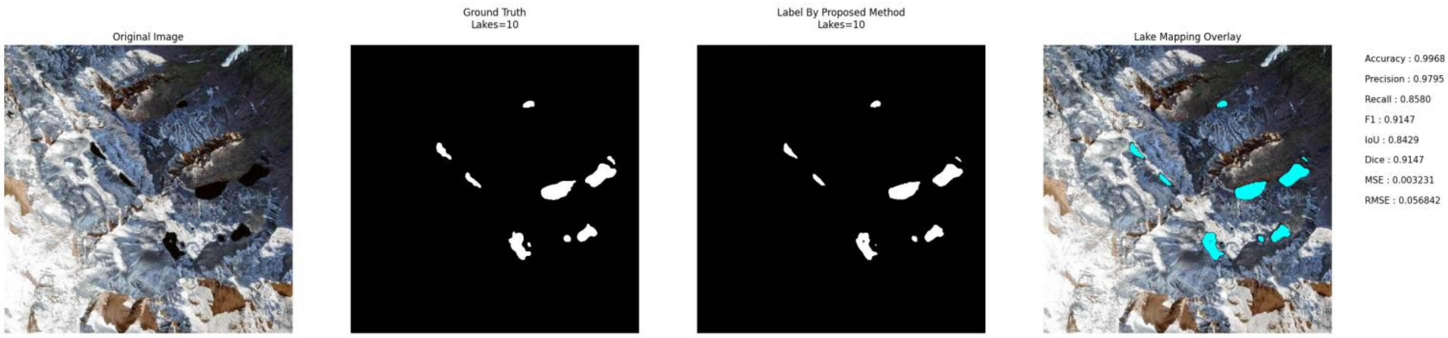


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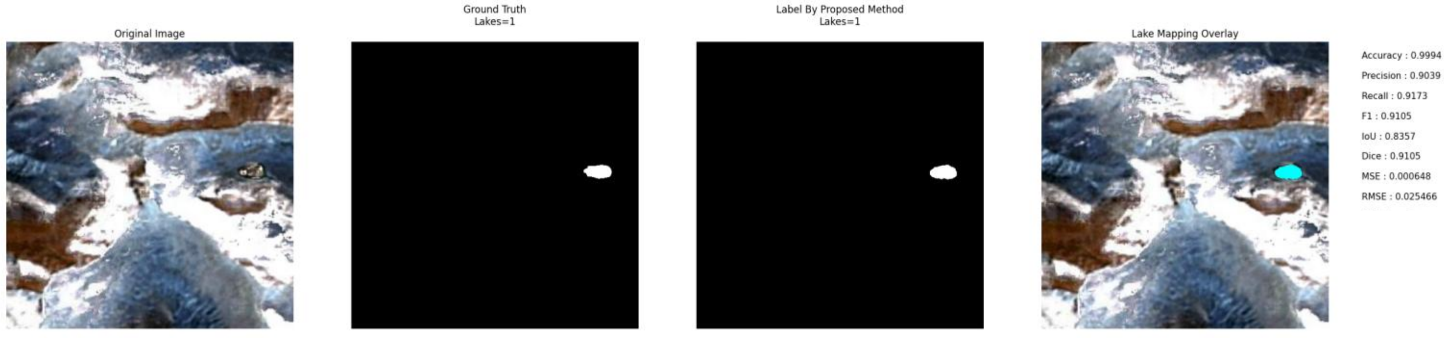


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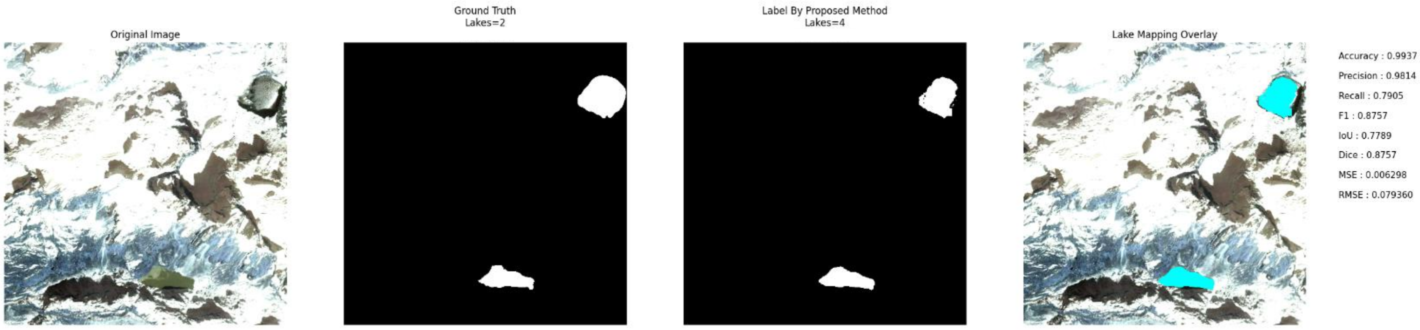


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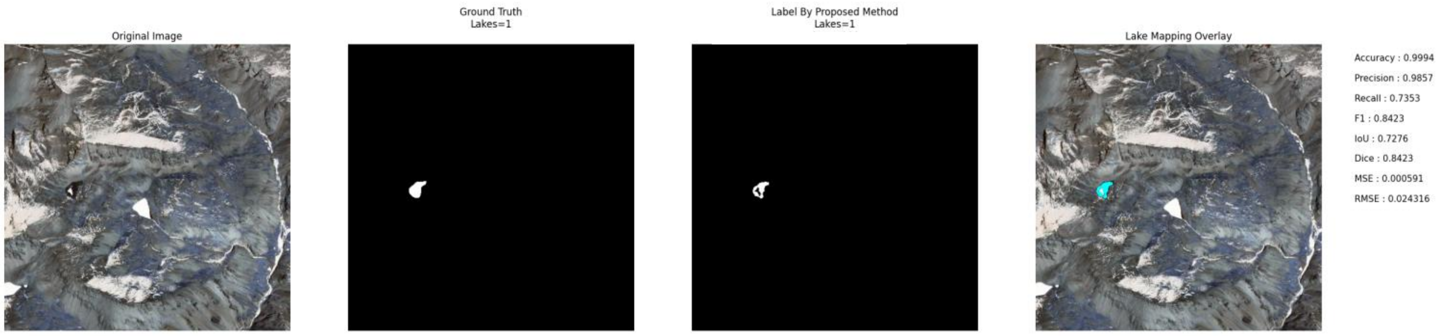



Image No: 1600

## 5. Moraine-Dammed Lakes

**Description:** Output images of moraine-dammed glacial lakes.

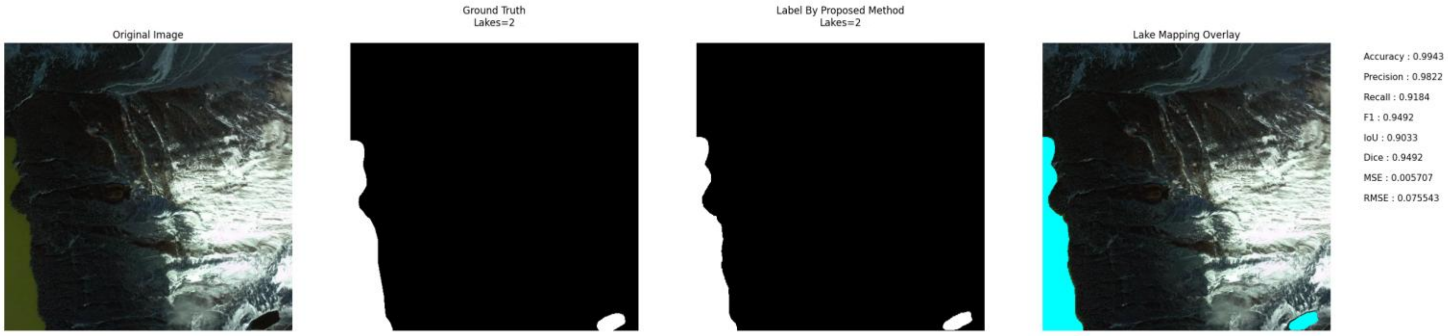
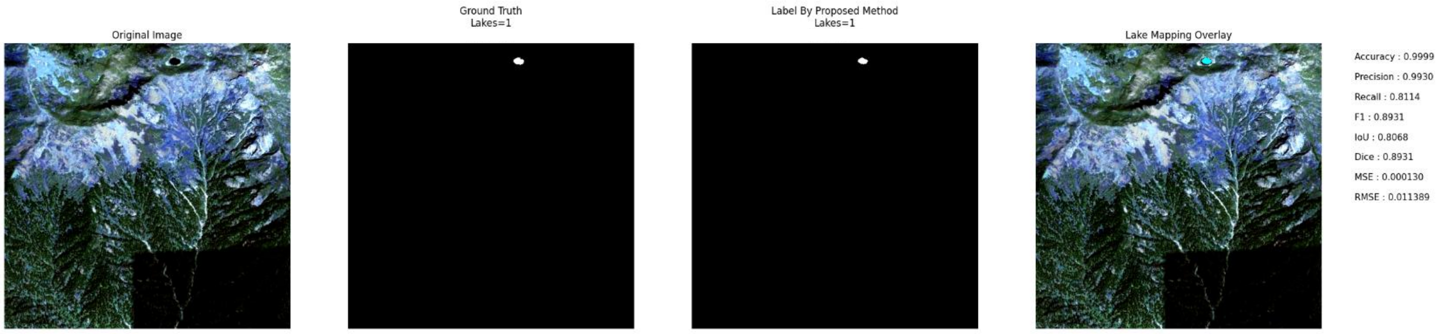



Image No: 5

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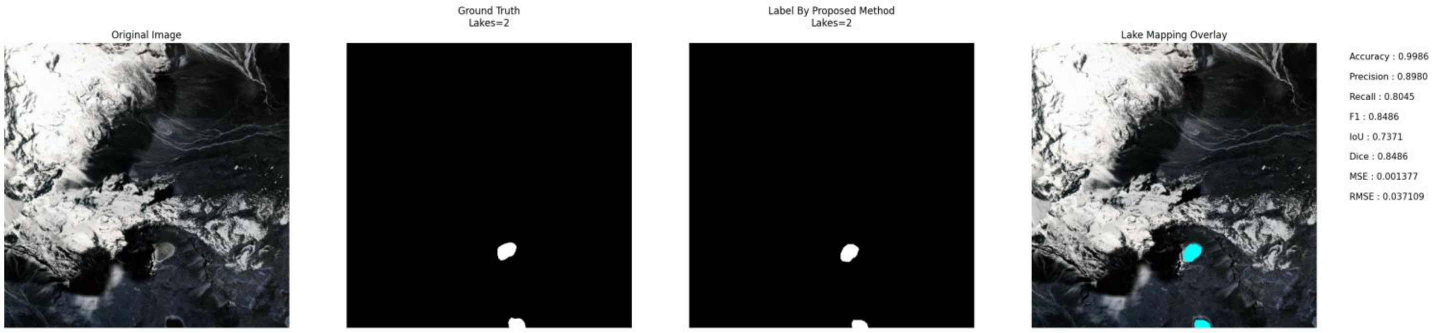


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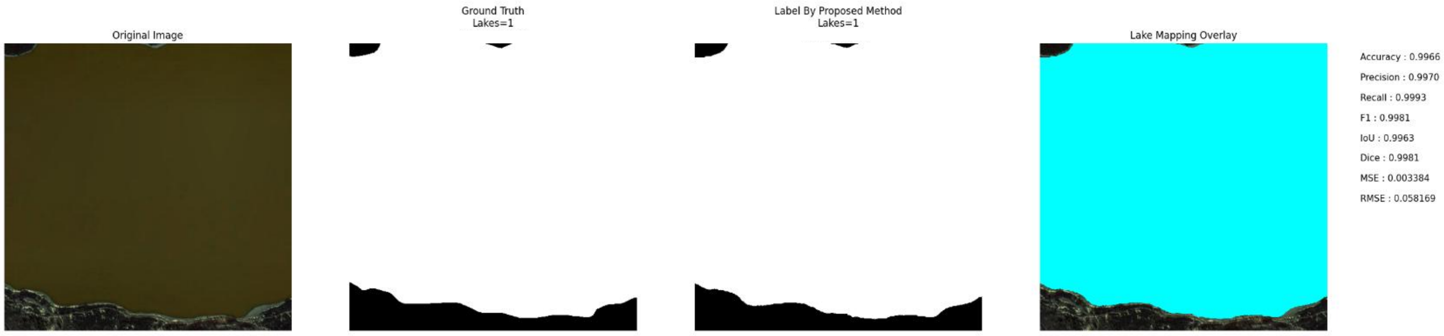


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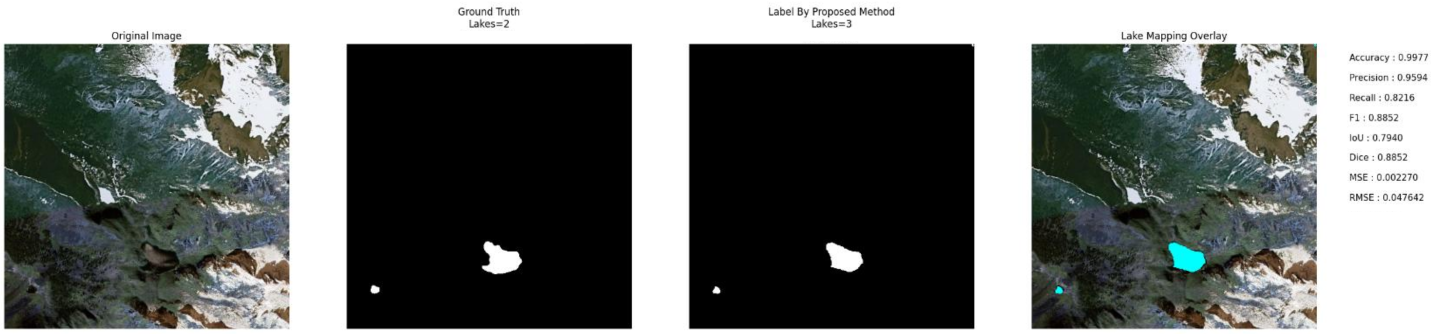


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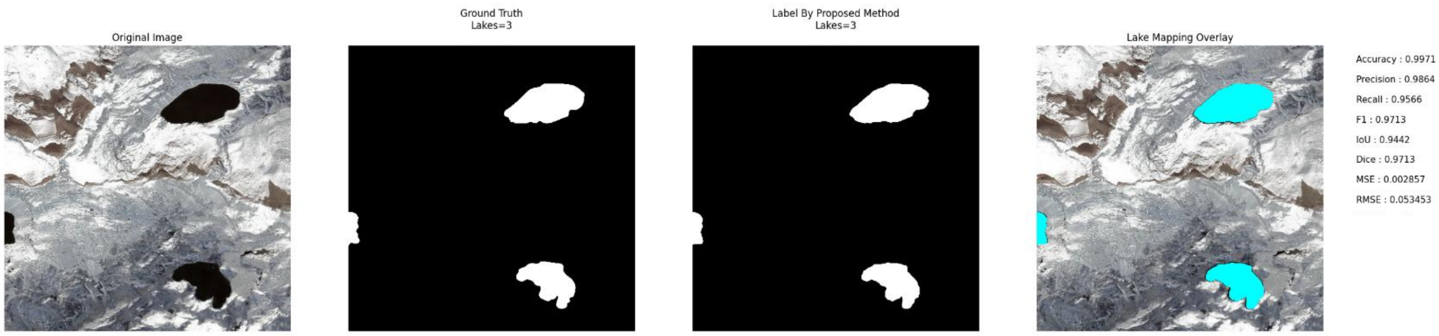


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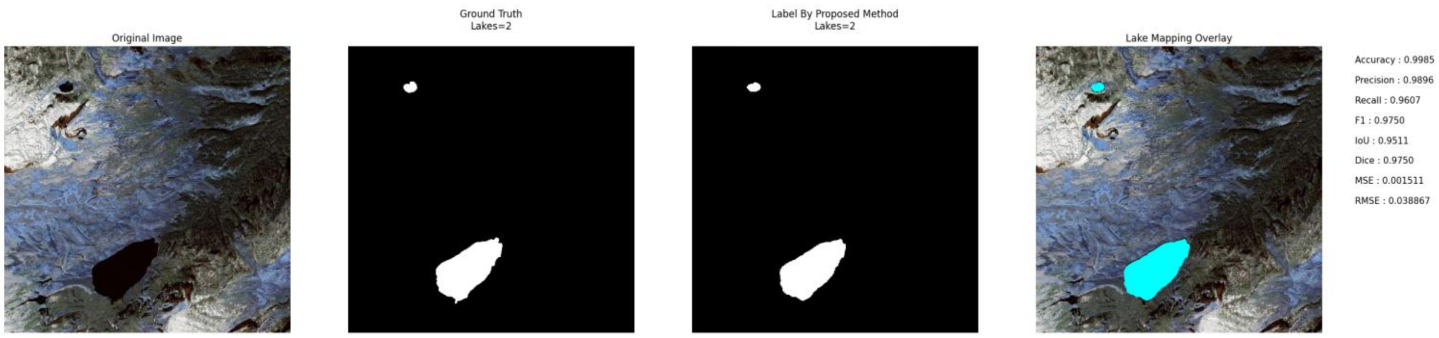


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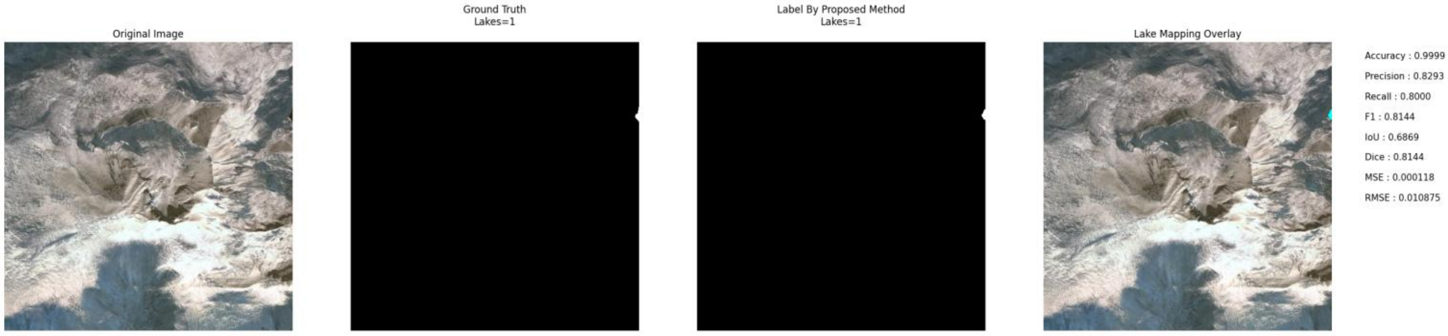


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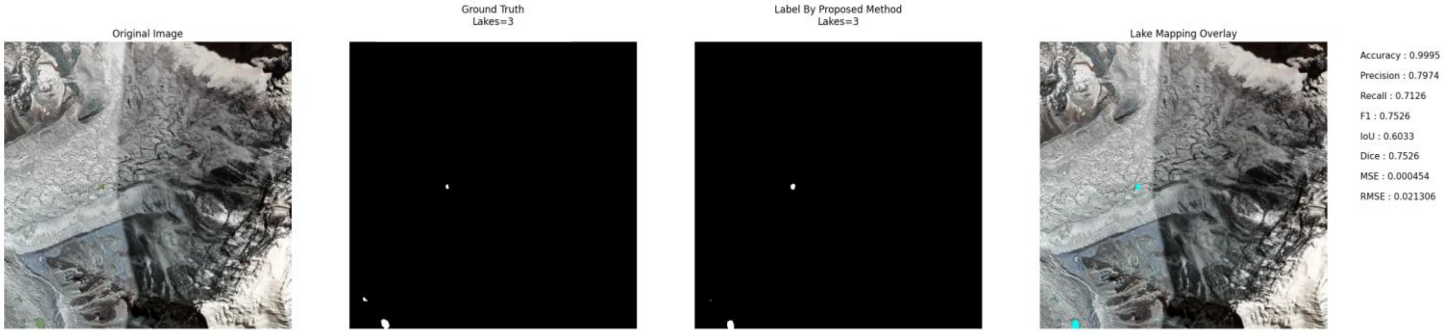


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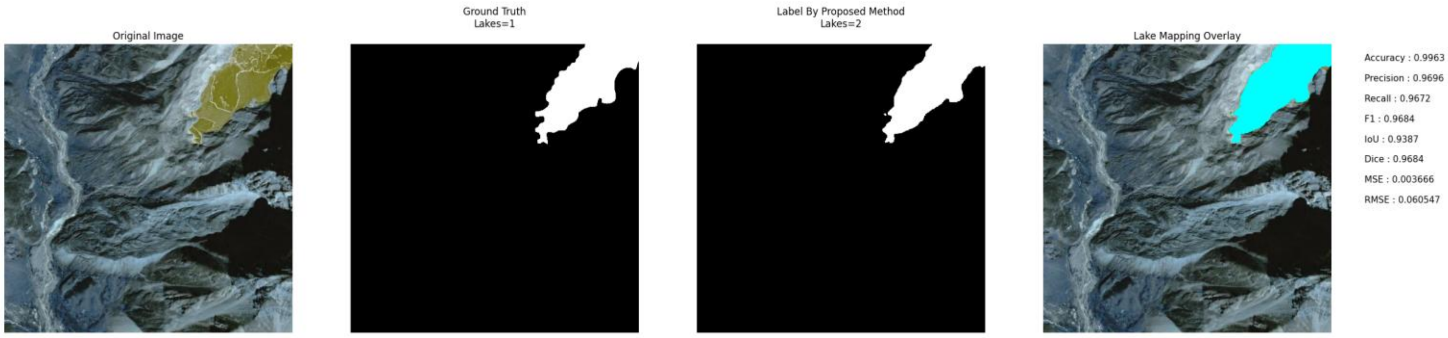
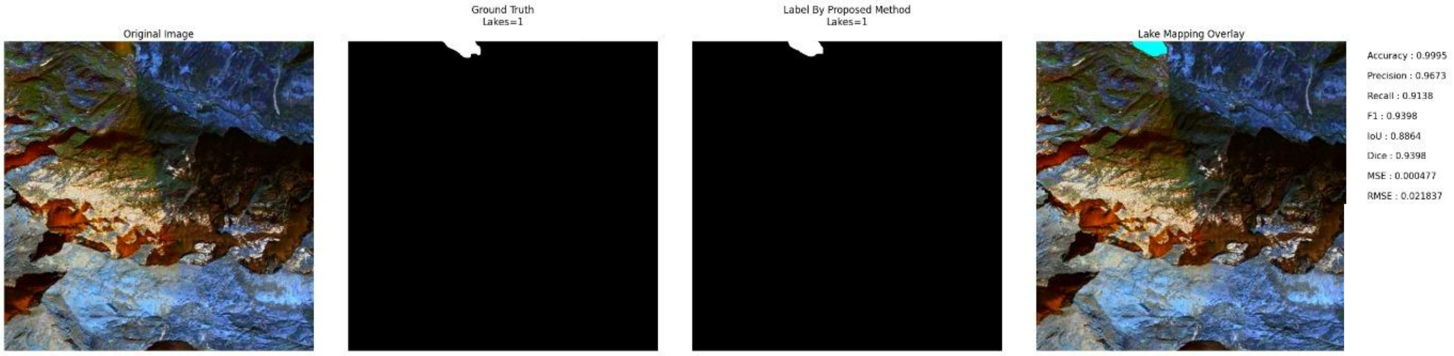



Image No: 50

# 6. Varying Turbidity

**Description:** Output images containing lakes with different turbidity levels.

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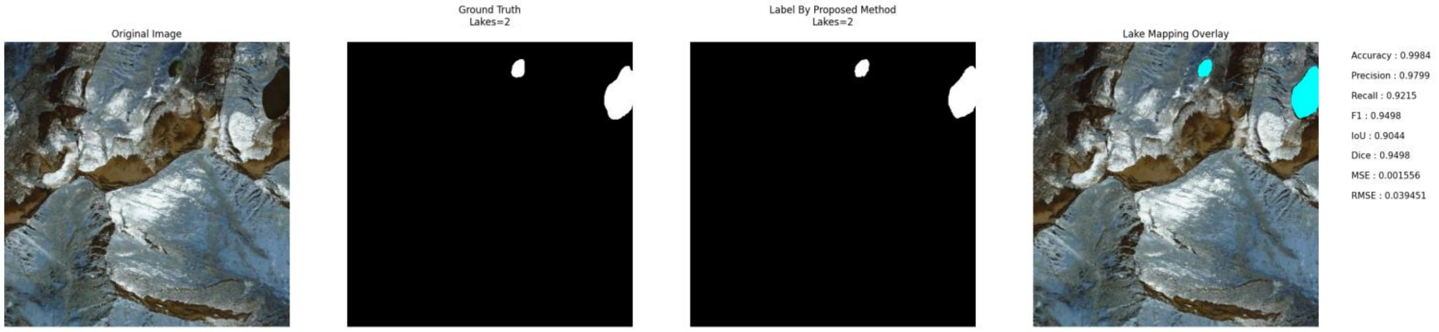


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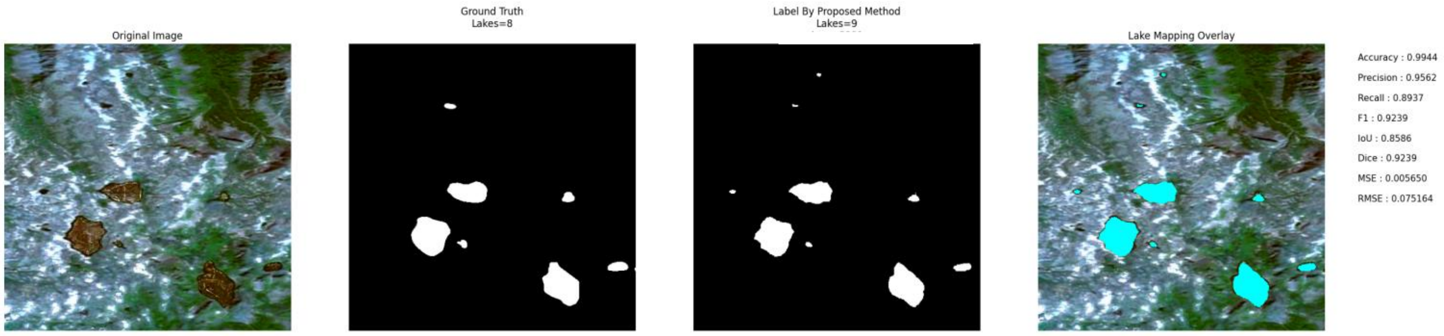


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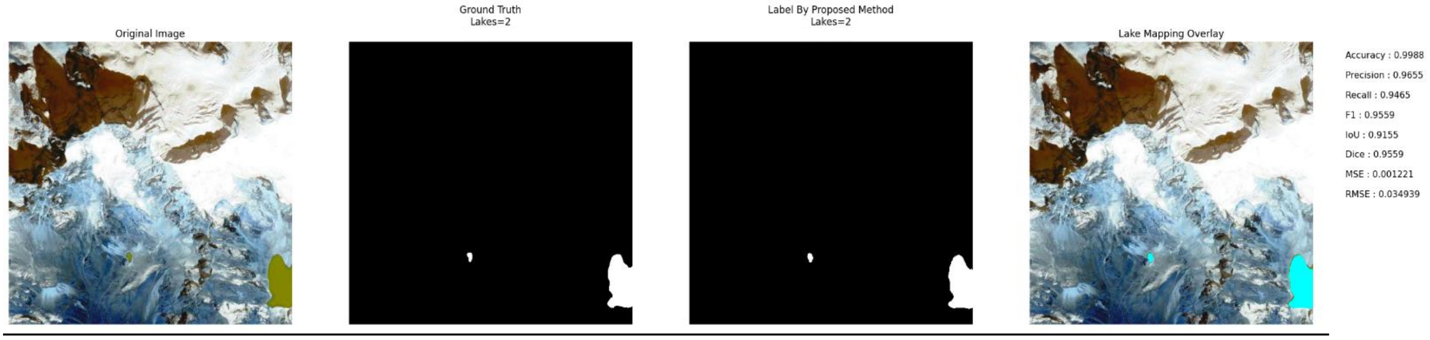


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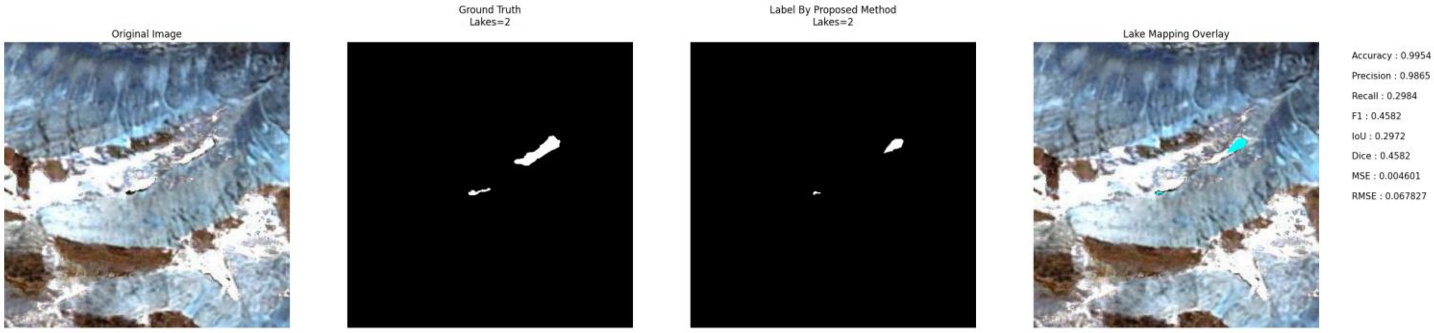


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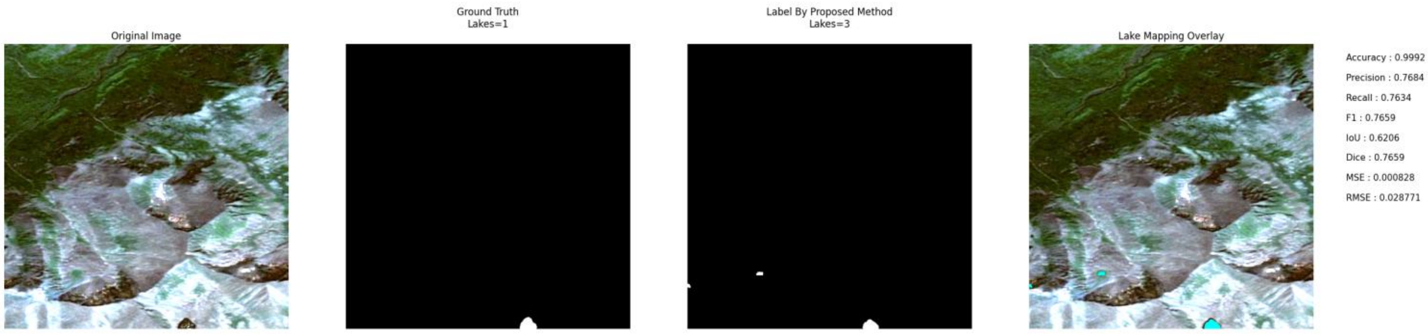


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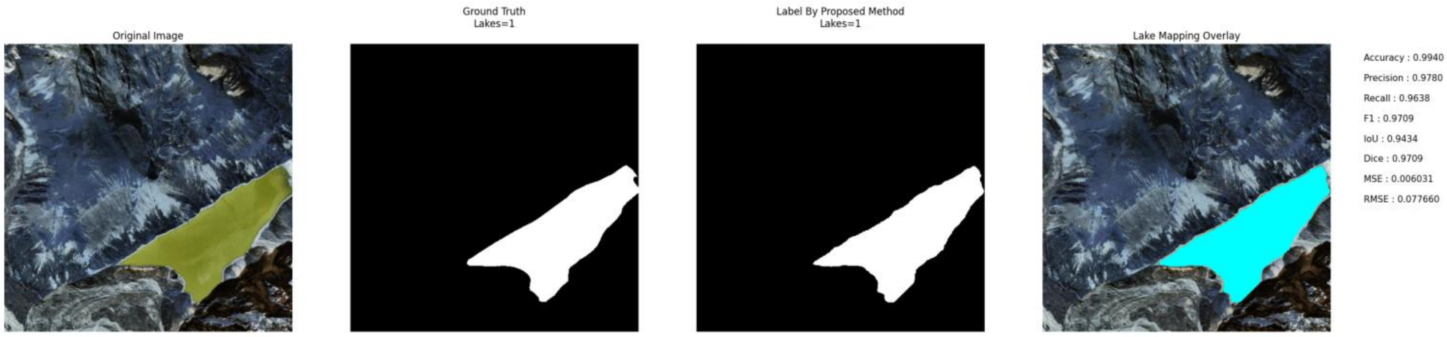


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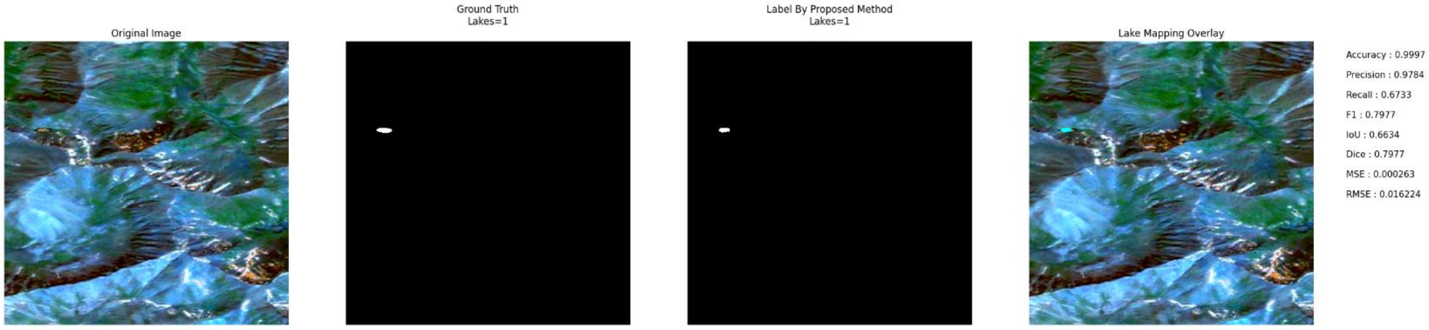


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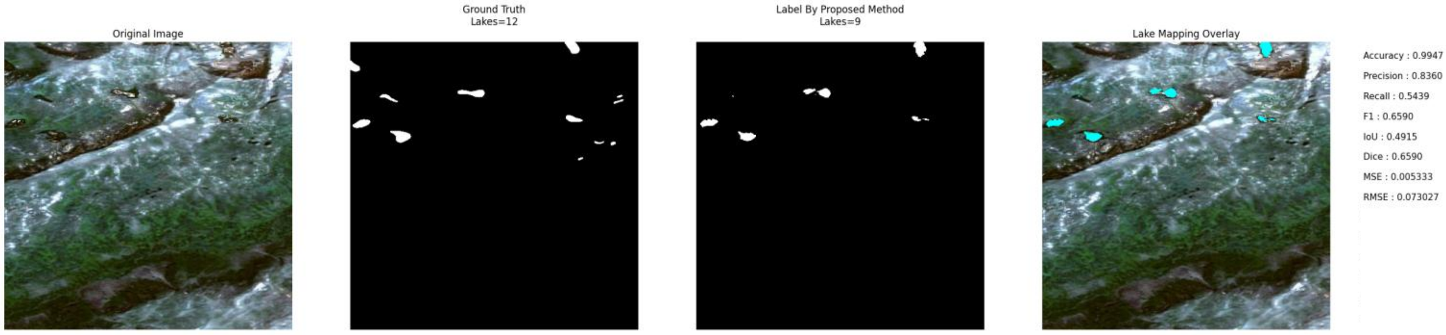


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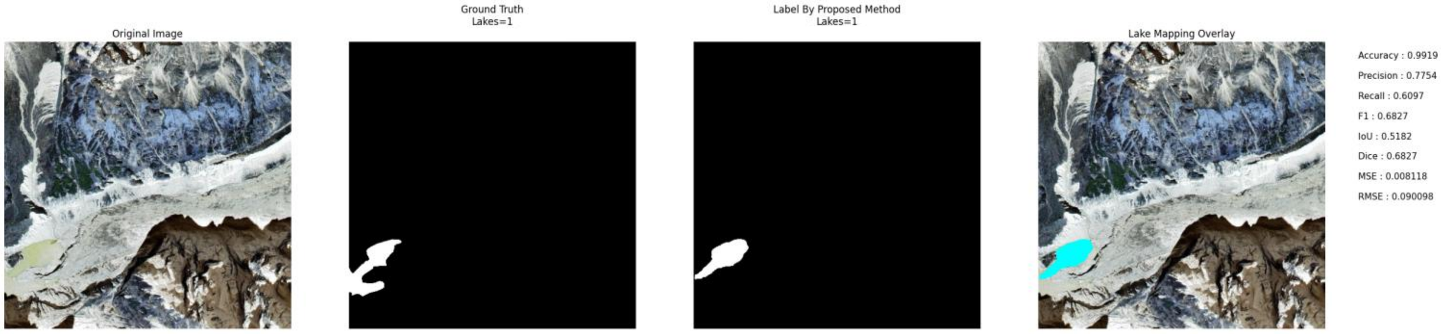


Image No: 1882

## Conclusion

The generated outputs demonstrate the effectiveness of the proposed framework in detecting glacial lakes under various challenging environmental conditions, including cloud cover, snow cover, terrain shadows, debris-covered regions, moraine-dammed lakes, and varying turbidity levels. The model exhibited strong generalization capability across diverse glacial landscapes and successfully differentiated glacial lakes from visually similar features such as snow, shadows, and debris-covered surfaces. Validation results indicate a high detection accuracy of **97.96%** with a low **Mean Squared Error (MSE) of 0.0165**, highlighting the reliability and precision of the proposed approach. The consistent performance across all six classes demonstrates the robustness of the framework for large-scale glacial lake mapping and monitoring. These results suggest that the proposed model can serve as a valuable tool for Glacial Lake Outburst Flood (GLOF) risk assessment, environmental monitoring, and climate change adaptation strategies.